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CAPSTONE PROJECT

NOTEBOOK 26: STRUCTURED MULTI-LABEL BASELINE ON CANDIDATE 150 LABELS

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Purpose:

This notebook trains the first structured multi-label genre

prediction baseline using the 150 candidate metadata genres.

The goal is to:

1. Load the candidate multi-label master table
2. Align engineered audio features to the same tracks
3. Remove rows with no candidate labels
4. Train structured multi-label baseline models

5. Evaluate using proper multi-label metrics

6. Save outputs for the next modelling phase

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In [1]:

```
# =====
# 1. IMPORT LIBRARIES
# =====

import os
import joblib
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from sklearn.multiclass import OneVsRestClassifier
from sklearn.linear_model import SGDClassifier
from sklearn.metrics import (
    f1_score,
    hamming_loss,
    accuracy_score,
    precision_score,
    recall_score
)
```

In [2]:

```
# =====
# 2. LOAD MULTI-LABEL MASTER TABLE AND FEATURES
# =====

master_df = pd.read_csv("../data/processed/multilabel_candidate_master_table.csv")

features = pd.read_csv(
    "../data/raw/metadata/features.csv",
    header=[0, 1, 2],
    index_col=0
)

features.index = features.index.astype(int)

print("Candidate multi-label master shape:", master_df.shape)
print("Features shape:", features.shape)
```

```
display(master_df.head())
```

Candidate multi-label master shape: (81574, 157)

Features shape: (106574, 518)

	track_id	split	subset	genre_top	title	audio_path	a
0	20	training	large	NaN	Spiritual Level	../data/raw/audio/fma_large\000\000020.mp3	
1	26	training	large	NaN	Where is your Love?	../data/raw/audio/fma_large\000\000026.mp3	
2	30	training	large	NaN	Too Happy	../data/raw/audio/fma_large\000\000030.mp3	
3	46	training	large	NaN	Yosemite	../data/raw/audio/fma_large\000\000046.mp3	
4	48	training	large	NaN	Light of Light	../data/raw/audio/fma_large\000\000048.mp3	

5 rows × 157 columns

In [3]:

```
# =====
# 3. IDENTIFY LABEL COLUMNS
# =====

label_cols = [
    col for col in master_df.columns
    if col.startswith("genre_") and col.replace("genre_", "").isdigit()
]

print("Number of candidate label columns:", len(label_cols))
print("First 10 label columns:", label_cols[:10])
print("\nNon-label genre-like columns:")
print([col for col in master_df.columns if col.startswith("genre_") and col not in
label_cols])
```

Number of candidate label columns: 150

First 10 label columns: ['genre_1', 'genre_2', 'genre_3', 'genre_4', 'genre_5', 'genre_6', 'genre_7', 'genre_8', 'genre_9', 'genre_10']

Non-label genre-like columns:
['genre_top']

In [4]:

```
# =====
# 4. REMOVE TRACKS WITH NO CANDIDATE LABELS
# =====
```

```

master_df["candidate_label_count"] = master_df[label_cols].sum(axis=1)

usable_df = master_df[master_df["candidate_label_count"] >
0].copy().reset_index(drop=True)

print("Original candidate master shape:", master_df.shape)
print("Usable candidate multi-label shape:", usable_df.shape)

print("\nSplit distribution after filtering:")
print(usable_df["split"].value_counts())

print("\nCandidate label count summary:")
print(usable_df["candidate_label_count"].describe())

```

Original candidate master shape: (81574, 158)

Usable candidate multi-label shape: (79343, 158)

Split distribution after filtering:

```

split
training      62452
test          8548
validation    8343
Name: count, dtype: int64

```

Candidate label count summary:

```

count      79343.000000
mean        3.679329
std         1.735141
min         1.000000
25%         3.000000
50%         4.000000
75%         5.000000
max         31.000000
Name: candidate_label_count, dtype: float64

```

In [5]:

```

# =====
# 5. ALIGN FEATURES TO USABLE TRACKS
# =====

usable_df["track_id"] = usable_df["track_id"].astype(int)

track_ids = usable_df["track_id"].tolist()
features_aligned = features.loc[track_ids].copy()

features_aligned = features_aligned.reset_index(drop=True)
usable_df = usable_df.reset_index(drop=True)

print("Aligned features shape:", features_aligned.shape)
print("Usable dataframe shape:", usable_df.shape)

```

Aligned features shape: (79343, 518)

Usable dataframe shape: (79343, 158)

In [6]:

```
# =====
# 6. FLATTEN FEATURE COLUMNS
# =====

features_aligned.columns = [
    "_".join([str(level) for level in col]).strip()
    for col in features_aligned.columns.to_flat_index()
]

print("Flattened features shape:", features_aligned.shape)
display(features_aligned.head())
```

Flattened features shape: (79343, 518)

	chroma_cens_kurtosis_01	chroma_cens_kurtosis_02	chroma_cens_kurtosis_03	chroma_cens_ku
0	-0.193837	-0.198527	0.201546	
1	-0.699535	-0.684158	0.048825	
2	-0.721487	-0.848560	0.890904	
3	-0.119708	-0.858814	2.362546	
4	-1.054053	0.932339	0.528064	

5 rows × 518 columns

In [7]:

```
# =====
# 7. PREPARE X AND Y
# =====

X = features_aligned.copy()
X = X.select_dtypes(include=["number"])
X = X.replace([np.inf, -np.inf], np.nan)
X = X.fillna(X.mean())
X = X.astype(np.float32)

Y = usable_df[label_cols].copy().astype(np.uint8)

print("X shape:", X.shape)
print("Y shape:", Y.shape)

print("\nAverage labels per track:", Y.sum(axis=1).mean())
print("Max labels on a track:", Y.sum(axis=1).max())
```

X shape: (79343, 518)

Y shape: (79343, 150)

Average labels per track: 3.67932898932483

Max labels on a track: 31

In [8]:

```
# =====
# 8A. CREATE TRAIN / VALIDATION / TEST SPLITS
# =====

split = usable_df["split"].astype(str)

train_mask = (split == "training").values
val_mask = (split == "validation").values
test_mask = (split == "test").values

X_train = X.loc[train_mask].copy()
Y_train = Y.loc[train_mask].copy()

X_val = X.loc[val_mask].copy()
Y_val = Y.loc[val_mask].copy()

X_test = X.loc[test_mask].copy()
Y_test = Y.loc[test_mask].copy()

print("Train:", X_train.shape, Y_train.shape)
print("Validation:", X_val.shape, Y_val.shape)
print("Test:", X_test.shape, Y_test.shape)
```

Train: (62452, 518) (62452, 150)

Validation: (8343, 518) (8343, 150)

Test: (8548, 518) (8548, 150)

In [9]:

```
# =====
# 8B. SAMPLE THE SPLITS FOR A SAFE BASELINE
# =====

SEED = 42

MAX_TRAIN_ROWS = 20000
MAX_VAL_ROWS = 5000
MAX_TEST_ROWS = 5000

if len(X_train) > MAX_TRAIN_ROWS:
    train_idx = X_train.sample(n=MAX_TRAIN_ROWS, random_state=SEED).index
    X_train = X_train.loc[train_idx].copy()
    Y_train = Y_train.loc[train_idx].copy()

if len(X_val) > MAX_VAL_ROWS:
    val_idx = X_val.sample(n=MAX_VAL_ROWS, random_state=SEED).index
    X_val = X_val.loc[val_idx].copy()
    Y_val = Y_val.loc[val_idx].copy()
```

```

if len(X_test) > MAX_TEST_ROWS:
    test_idx = X_test.sample(n=MAX_TEST_ROWS, random_state=SEED).index
    X_test = X_test.loc[test_idx].copy()
    Y_test = Y_test.loc[test_idx].copy()

print("Sampled safe baseline sizes:")
print("Train:", X_train.shape, Y_train.shape)
print("Validation:", X_val.shape, Y_val.shape)
print("Test:", X_test.shape, Y_test.shape)

```

Sampled safe baseline sizes:
 Train: (20000, 518) (20000, 150)
 Validation: (5000, 518) (5000, 150)
 Test: (5000, 518) (5000, 150)

In [10]:

```

# =====
# 9. SCALE FEATURES
# =====

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train).astype(np.float32)
X_val_scaled = scaler.transform(X_val).astype(np.float32)
X_test_scaled = scaler.transform(X_test).astype(np.float32)

print("Scaling complete.")

```

Scaling complete.

In [11]:

```

# =====
# 10. DEFINE EVALUATION HELPERS
# =====

def evaluate_multilabel(y_true, y_pred, name="Model"):
    results = {
        "Model": name,
        "Micro F1": f1_score(y_true, y_pred, average="micro", zero_division=0),
        "Macro F1": f1_score(y_true, y_pred, average="macro", zero_division=0),
        "Samples F1": f1_score(y_true, y_pred, average="samples", zero_division=0),
        "Micro Precision": precision_score(y_true, y_pred, average="micro",
zero_division=0),
        "Micro Recall": recall_score(y_true, y_pred, average="micro",
zero_division=0),
        "Hamming Loss": hamming_loss(y_true, y_pred),
        "Subset Accuracy": accuracy_score(y_true, y_pred)
    }
    return results

def get_score_matrix(model, X_input):
    if hasattr(model, "predict_proba"):

```

```

        scores = model.predict_proba(X_input)
    elif hasattr(model, "decision_function"):
        scores = model.decision_function(X_input)
    else:
        raise ValueError("Model does not support predict_proba or
decision_function.")
    return np.asarray(scores)

def precision_recall_at_k(y_true, score_matrix, k=3):
    y_true = np.asarray(y_true)
    topk_idx = np.argsort(-score_matrix, axis=1)[: , :k]

    precision_vals = []
    recall_vals = []

    for i in range(y_true.shape[0]):
        true_idx = np.where(y_true[i] == 1)[0]
        if len(true_idx) == 0:
            continue

        pred_idx = topk_idx[i]
        hits = len(set(pred_idx).intersection(set(true_idx)))

        precision_vals.append(hits / k)
        recall_vals.append(hits / len(true_idx))

    return float(np.mean(precision_vals)), float(np.mean(recall_vals))

```

In [12]:

```

# =====
# 11. TRAIN ONE-VS-REST SGD LOGISTIC
# =====

ovr_sgd_log = OneVsRestClassifier(
    SGDClassifier(
        loss="log_loss",
        alpha=1e-4,
        max_iter=20,
        tol=1e-3,
        class_weight="balanced",
        random_state=42
    ),
    n_jobs=1
)

ovr_sgd_log.fit(X_train_scaled, Y_train)

Y_val_pred_log = ovr_sgd_log.predict(X_val_scaled)
val_scores_log = get_score_matrix(ovr_sgd_log, X_val_scaled)

log_results = evaluate_multilabel(Y_val, Y_val_pred_log, name="OVR SGD Logistic")

p_at_3_log, r_at_3_log = precision_recall_at_k(Y_val.values, val_scores_log, k=3)
p_at_5_log, r_at_5_log = precision_recall_at_k(Y_val.values, val_scores_log, k=5)

```

```
log_results["Precision@3"] = p_at_3_log
log_results["Recall@3"] = r_at_3_log
log_results["Precision@5"] = p_at_5_log
log_results["Recall@5"] = r_at_5_log

print("Finished OVR SGD Logistic.")
print(log_results)
```

```

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packages\sklearn\linear_model\_stochastic_gradient.py:738: ConvergenceWarning:
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[illegible]

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    warnings.warn(
Finished OVR SGD Logistic.
{'Model': 'OVR SGD Logistic', 'Micro F1': 0.15931854766589162, 'Macro F1':
0.08013485821520762, 'Samples F1': 0.15868185647935196, 'Micro Precision':
0.09322263018811976, 'Micro Recall': 0.547508067407673, 'Hamming Loss': 0.150408,
'Subset Accuracy': 0.0, 'Precision@3': 0.05793333333333332, 'Recall@3':
0.045224435786435785, 'Precision@5': 0.05536, 'Recall@5': 0.0719961818181818}
```

In [13]:

```
# =====  
# 12. TRAIN ONE-VS-REST SGD HINGE  
# =====  
  
ovr_sgd_hinge = OneVsRestClassifier(  
    SGDClassifier(  
        loss="hinge",  
        alpha=1e-4,  
        max_iter=20,  
        tol=1e-3,  
        class_weight="balanced",  
        random_state=42  
    ),  
    n_jobs=1
```

```
)

ovr_sgd_hinge.fit(X_train_scaled, Y_train)

Y_val_pred_hinge = ovr_sgd_hinge.predict(X_val_scaled)
val_scores_hinge = get_score_matrix(ovr_sgd_hinge, X_val_scaled)

hinge_results = evaluate_multilabel(Y_val, Y_val_pred_hinge, name="OVR SGD Hinge")

p_at_3_hinge, r_at_3_hinge = precision_recall_at_k(Y_val.values, val_scores_hinge,
k=3)
p_at_5_hinge, r_at_5_hinge = precision_recall_at_k(Y_val.values, val_scores_hinge,
k=5)

hinge_results["Precision@3"] = p_at_3_hinge
hinge_results["Recall@3"] = r_at_3_hinge
hinge_results["Precision@5"] = p_at_5_hinge
hinge_results["Recall@5"] = r_at_5_hinge

print("Finished OVR SGD Hinge.")
print(hinge_results)
```

```

c:\Users\jdev\AppData\Local\Programs\Python\Python39\lib\site-
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[illegible]

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warnings.warn(
Finished OVR SGD Hinge.
{'Model': 'OVR SGD Hinge', 'Micro F1': 0.16272981674841272, 'Macro F1':
0.07915137082646072, 'Samples F1': 0.1630859834765631, 'Micro Precision':
0.09537161953636837, 'Micro Recall': 0.5540132151820929, 'Hamming Loss': 0.1484,
'Subset Accuracy': 0.0, 'Precision@3': 0.024933333333333333, 'Recall@3':
0.019416111111111111, 'Precision@5': 0.03, 'Recall@5': 0.03788651515151515}
```

In [14]:

```
# =====
# 13. VALIDATION COMPARISON SUMMARY
```

```
# =====

validation_results_df = pd.DataFrame([log_results, hinge_results])

print("Validation comparison summary:")
display(validation_results_df)
```

Validation comparison summary:

	Model	Micro F1	Macro F1	Samples F1	Micro Precision	Micro Recall	Hamming Loss	Subset Accuracy	Precision@
0	OVR SGD Logistic	0.159319	0.080135	0.158682	0.093223	0.547508	0.150408	0.0	0.05793
1	OVR SGD Hinge	0.162730	0.079151	0.163086	0.095372	0.554013	0.148400	0.0	0.02493



In [15]:

```
# =====
# 14. SELECT BEST MODEL BASED ON MICRO F1
# =====

best_model_name = validation_results_df.sort_values("Micro F1",
ascending=False).iloc[0]["Model"]

print("Best multi-label structured model:", best_model_name)

if best_model_name == "OVR SGD Logistic":
    best_model = ovr_sgd_log
else:
    best_model = ovr_sgd_hinge
```

Best multi-label structured model: OVR SGD Hinge

In [16]:

```
# =====
# 15. FINAL TEST EVALUATION
# =====

Y_test_pred = best_model.predict(X_test_scaled)
test_scores = get_score_matrix(best_model, X_test_scaled)

test_results = evaluate_multilabel(Y_test, Y_test_pred, name=best_model_name)

p_at_3_test, r_at_3_test = precision_recall_at_k(Y_test.values, test_scores, k=3)
p_at_5_test, r_at_5_test = precision_recall_at_k(Y_test.values, test_scores, k=5)

test_results["Precision@3"] = p_at_3_test
test_results["Recall@3"] = r_at_3_test
```

```
test_results["Precision@5"] = p_at_5_test
test_results["Recall@5"] = r_at_5_test

print("Final test results:")
print(test_results)
```

Final test results:

```
{'Model': 'OVR SGD Hinge', 'Micro F1': 0.1611562096957742, 'Macro F1':
0.0836051945282357, 'Samples F1': 0.161468435208297, 'Micro Precision':
0.0937009263343626, 'Micro Recall': 0.5753521126760563, 'Hamming Loss': 0.147424,
'Subset Accuracy': 0.0, 'Precision@3': 0.02486666666666665, 'Recall@3':
0.019935714285714282, 'Precision@5': 0.03196, 'Recall@5': 0.042197460317460314}
```

In [17]:

```
# =====
# 16. SAVE MODELS AND OUTPUTS
# =====

os.makedirs("../models", exist_ok=True)
os.makedirs("../data/processed", exist_ok=True)

joblib.dump(best_model, "../models/best_structured_multilabel_candidate150.joblib")
joblib.dump(scaler, "../models/structured_multilabel_candidate150_scaler.joblib")

validation_results_df.to_csv(
    "../data/processed/structured_multilabel_candidate150_validation_results.csv",
    index=False
)

pd.DataFrame([test_results]).to_csv(
    "../data/processed/structured_multilabel_candidate150_test_results.csv",
    index=False
)

# Save the correct validation score matrix based on the selected best model
if best_model_name == "OVR SGD Logistic":
    best_val_scores = val_scores_log
else:
    best_val_scores = val_scores_hinge

np.save("../data/processed/structured_multilabel_candidate150_val_scores.npy",
best_val_scores)
np.save("../data/processed/structured_multilabel_candidate150_test_scores.npy",
test_scores)

np.save("../data/processed/structured_multilabel_candidate150_y_val.npy",
Y_val.values)
np.save("../data/processed/structured_multilabel_candidate150_y_test.npy",
Y_test.values)
np.save("../data/processed/structured_multilabel_candidate150_label_columns.npy",
np.array(label_cols))

with open("../data/processed/best_structured_multilabel_candidate150_name.txt",
"w") as f:
    f.write(best_model_name)
```

```
print("Saved multi-label structured baseline outputs.")
```

Saved multi-label structured baseline outputs.

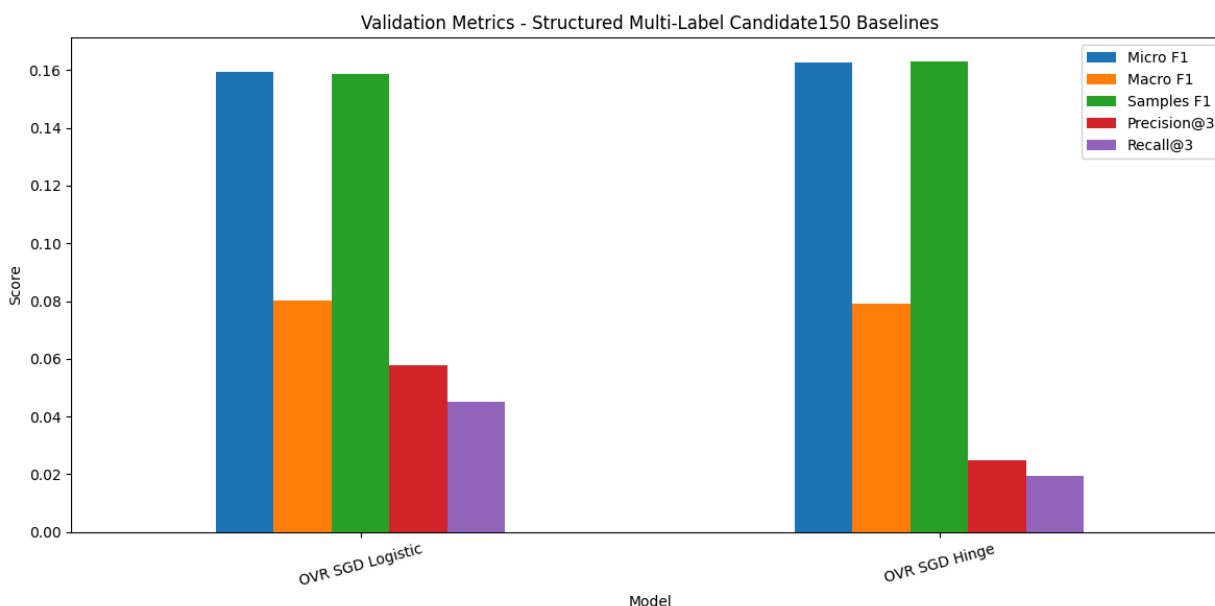
In [18]:

```
# =====
# 17. PLOT MAIN VALIDATION METRICS
# =====

plot_cols = ["Micro F1", "Macro F1", "Samples F1", "Precision@3", "Recall@3"]

plot_df = validation_results_df.set_index("Model")[plot_cols]

ax = plot_df.plot(kind="bar", figsize=(12, 6))
ax.set_title("Validation Metrics - Structured Multi-Label Candidate150 Baselines")
ax.set_ylabel("Score")
plt.xticks(rotation=15)
plt.tight_layout()
plt.show()
```



In [19]:

```
# =====
# 18. INTERPRETATION NOTES
# =====

print("1. This notebook establishes the first structured multi-label baseline on the 150 candidate labels.")
print("2. Rows with no candidate labels were removed so the model is trained only on usable candidate-label examples.")
print("3. Multi-label evaluation uses F1, Hamming loss, subset accuracy, and top-k retrieval-style metrics.")
print("4. The saved outputs can support future comparison with audio multi-label
```

```
and hybrid multi-label models.")  
print("5. The next logical step is an audio multi-label baseline or a MongoDB-  
driven prediction/retrieval workflow.")
```

1. This notebook establishes the first structured multi-label baseline on the 150 candidate labels.
2. Rows with no candidate labels were removed so the model is trained only on usable candidate-label examples.
3. Multi-label evaluation uses F1, Hamming loss, subset accuracy, and top-k retrieval-style metrics.
4. The saved outputs can support future comparison with audio multi-label and hybrid multi-label models.
5. The next logical step is an audio multi-label baseline or a MongoDB-driven prediction/retrieval workflow.